Proposition 8: fixes

Conditional work plan | latest source rechecked on 12 October 2026

Items 1-6 are resolved. Item 9 is partly corrected; the remaining items are open. Finish the proof while temporarily assuming (R), then prove (R) last.

(R) For every fixed $A > 0$, $a_N(A) \geq 0$ tends to zero and is independent of j, with

$$\|r_{N,j}\|_2 \leq |\delta| a_N(A) \text{ for every } j \in I_N(A).$$

Labels: NOW = finish under (R); RED = proves (R), defer; BOTH = contains shared or split work. The full proposition remains conditional until (R) is proved.

Notation: $a = h_{min}$, $b = h_{max}$, $\delta = \delta_{ij}^{r,N}$, $\varepsilon = \varepsilon_N$. Q_r denotes the empirical energy. No thesis files were changed.

1. State the fixed-sample scope. [RESOLVED]

Fixed data, fixed positive endpoints, and a fixed grid exponent are specified. Sample size stays fixed as $N \to \infty$.

Source: bandwidth-selection chapter, lines 767-774.

2. Correct the divergence domain. [RESOLVED]

D is defined on $P_2 \times P_2$, where $P_2 = P \cap L^2(\mathbb{R})$.

Source: bandwidth-selection chapter, line 768.

3. Repair the grid-spacing lemma. [RESOLVED]

Fixed endpoints, the powered grid identity, $c_N = O(\delta)$, the $o(N)$ condition, and the $O(\delta^2)$ remainder are correct.

Source: bandwidth-selection chapter, lines 542-564.

4. Finish the integer count. [RESOLVED]

Both endpoints are shifted correctly; $\#I_N(A) \leq 2\lfloor A/\varepsilon \rfloor + 1 \leq 2A/\varepsilon + 1$.

Source: bandwidth-selection chapter, line 780.

5. Finish the Fourier lower bound. [RESOLVED]

Absolute values, ordered limits, the squared exponential, $c = a^{4-2r}/(4\pi) \int g$, and positivity are correct.

Source: bandwidth-selection chapter, lines 787-796.

6. Normalize the tail. [RESOLVED]

The ε factors, boundedness sentence, scaled limsup, and $\exp(-cA^2)/(cA)$ bound are correct.

Source: bandwidth-selection chapter, lines 800-806.

7. Correct remainder inequalities. [BOTH]

Now: in lines 824-825, put absolute values around δ and the inner product before using the two norms. Use $\|r\|_2 \leq |\delta| a_N(A)$. **Later:** the ratio for u uses $|\tau_{N,j}|$ for nonzero increments; its uniform justification and the zero-increment case belong to the proof of (R).

$$|\rho_{N,j}| \leq |\delta| M \|r_{N,j}\|_2 + \tfrac{1}{2}\|r_{N,j}\|_2^2.$$

Source: bandwidth-selection chapter, lines 814; 822-828.

8. Prove the first red estimate last. [RED]

Defer the TODO before Lemma 7 and the proof of (R). A C^1 KDE map suffices: use the first-derivative integral formula, uniform continuity, and a uniform grid-spacing constant. A C^2 Taylor bound is an optional stronger route. While completing Lemma 7, fix the interval order for negative τ at line 745 and change the equality of the two suprema at line 746 to $\leq$.

Source: bandwidth-selection chapter, lines 724; 745-746; 809-816. Zeidler, §4.4, Corollary 5, p. 242 (C^1); §4.5, Theorem 4.C, pp. 243-244 (C^2).

Proposition 8: fixes

Open work under the red assumption | checklist continued

9. Correct the exponential justification. [PARTIAL / NOW]

The new estimate at line 848 is correct under (R), after item 7 and the derivative bound in item 10. Your current $b_N(A)$ excludes A^2, so the A^2 factors below are correct. Replace the reason at line 834 by this local estimate; $\lambda_N/N^2 \to 0$ supplies the mesh condition. Explicitly take the supremum in the conclusion.

$$b_N(A) = Ma_N(A)+\tfrac{1}{2}a_N(A)^2 \to 0;$$
$$\sup_{j\in I_N(A)} \lambda_N|\rho_{N,j}| \le A^2 b_N(A),$$
$$\sup_{j\in I_N(A)}|\eta_{N,j}| \le \exp(A^2 b_N(A))-1 \to 0.$$

Source: bandwidth-selection chapter, lines 826; 834; 848.

10. Justify M and the energy properties. [BOTH]

Now: establish continuity of $h \mapsto \partial_h \tilde{f}_h$ in L^2 on the fixed positive interval. Gaussian domination supplies this and the finite bound M used in item 7. Show Q_r is continuous and positive using $P_n(0) = 1$. **Also helps (R):** the same derivative-continuity check is the main analytic ingredient for the deferred C^1 proof.

$$Q_r(h) = h^{4-2r}/(4\pi) \int_{\mathbb{R}} |P_n(\omega)|^2 \omega^4 \exp(-h^2\omega^2)\, d\omega > 0.$$

Source: bandwidth-selection chapter, lines 358-366; 825; 851.

11. Complete the Riemann-sum setup. [NOW]

State $\varepsilon_N \to 0$, which follows from $\lambda_N/N^2 \to 0$ and $\Delta_{r,N} = O(1/N)$. For $v_{i_N} = h_{i_N}{}^r/r$ tending to an interior point, both endpoint distances multiplied by $\sqrt{\lambda_N}$ tend to infinity. Thus the fixed local window is not truncated. Restrict j explicitly to 1,...,N in $I_N(A)$.

Source: bandwidth-selection chapter, lines 778; 851-863.

12. Replace the final fixed-A equality. [NOW]

At line 867, the full row sum cannot be equated to the truncated integral for a fixed A. Put $L_A = \int_{-A}^{A} \exp(-t^2 Q_r(h_0))\, dt$. Use the local limit and nonnegative tail for the bounds below, then let $A \to \infty$.

$$L_A \le \liminf_N \varepsilon R_{i_N}{}^{(N)} \le \limsup_N \varepsilon R_{i_N}{}^{(N)}$$
$$\le L_A + 2\int_A^{\infty} \exp(-ct^2)\, dt.$$

Source: bandwidth-selection chapter, lines 865-871.

13. Make the notation consistent. [BOTH]

Now: use $\delta_{ij}^{r,N}$, $\rho_{N,j}$, $\eta_{N,j}$, and $S^{(N)}$ consistently in the argument after (R). Replace the undefined δ_N, $\delta_{N,j}$ and the typo $r_{N,j}$. **Later:** finish the u/τ notation in the construction of the remainder with item 8. Keep $b_N(A)$ as currently defined; do not absorb A^2 into it elsewhere.

Source: bandwidth-selection chapter, lines 809-848.

14. Qualify the interpretation and Perron claim. [NOW]

The row-sum interpretation can be finished after the conditional proof: it compares fixed interior limiting bandwidths. Use Q_r, or explain $\beta = 2(1-r)$; slow change refers to $v = h^r/r$. The functional measures squared speed. Perron-vector concentration in the heading is a separate result: assuming (R) does not prove it.

Source: bandwidth-selection chapter, lines 714; 875-884.

Work order: downstream parts of 7 and 13 → 10 and 11 → remaining sentence in 9 → 12 → row-sum interpretation in 14. Then return to 8 and the deferred parts of 7 and 13.

All line references: /Users/iljasukhanov/MASTERARBEIT/Abschlussarbeit_2021/LaTeX/chapters/5_bandwidth_selection.tex. Calculations and classifications are original reasoning documented in output/analysis/proposition_8_review.md.